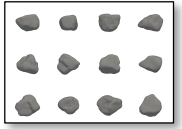


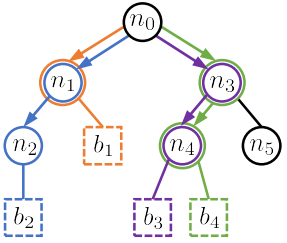
Scene $\mathcal{X}_k$ and target structure T



Inventory $\mathcal{I}_k$

State s_k

Batched selection



Parallel node expansion and evaluation

π_{init} π_{opt} π_{sim}

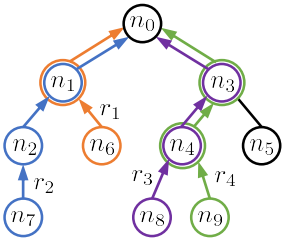
$b_1 : s(n_1) \longrightarrow \hat{a}_1 \longrightarrow a_1 \longrightarrow s(n_6), r_1$

$b_2 : s(n_2) \longrightarrow \hat{a}_2 \longrightarrow a_2 \longrightarrow s(n_7), r_2$

$b_3 : s(n_4) \longrightarrow \hat{a}_3 \longrightarrow a_3 \longrightarrow s(n_8), r_3$

$b_4 : s(n_4) \longrightarrow \hat{a}_4 \longrightarrow a_4 \longrightarrow s(n_9), r_4$

Backpropagation



Repeat while search budget remains

After search

Selected validated action a_k^*